Name: V Vishwa Teja ID number: 2520090033

Week 1

SESSION-1

Task 1: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Make file.

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill1.1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

CODE:

```
 GNU nano 8.7.1
#include <stdio.h>
int main() {
        printf("Hello, OSSP Skill Week 1!\n");
        printf("GCC compilation is working successfully.\n");
        return 0;
}
```

OUTPUT:

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill1.1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc skill1.1.c -o skill1.1
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./skill1.1
Hello, OSSP Skill Week 1!
GCC compilation is working successfully.
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

```
 GNU nano 8.7.1
CC = gcc
CFLAGS = -Wall -Wextra

all: main

main: main.c
$(CC) $(CFLAGS) main.c -o main

clean:
rm -f main
```

Verifying linux environment by uname-a:

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ uname -a
Linux LAPTOP-DID88SB3 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

Checking GCC C compiler:

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

Git version:

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ git --version
git version 2.53.0
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

Task 2: To Analyse process abstraction, execute fork(), understand exec() family, analyse parent-child relationships, inspect process tree, practice system call tracing.

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill1.2.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

CODE:

```
  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>
int main() {
        pid_t pid;
        printf("Parent Process Started\n");
        printf("Parent PID: %d\n", getpid());
        pid = fork();
        if (pid < 0) {
                perror("fork failed");
                return 1;
}
```

```c
        if (pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());
        printf("Executing ls using exec()\n");
        execlp("ls", "ls", "-l", NULL);
        perror("exec failed");
        exit(1);
}
else {
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
        printf("Parent waiting for child ... \n");
        wait(NULL);
        printf("Child process completed.\n");
}
        return 0;
}
```

OUTPUT:

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill1.2.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc skill1.2.c -o skill1.2
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./skill1.2
Parent Process Started
Parent PID: 867

Parent Process
Parent PID: 867
Child PID: 868
Parent waiting for child ...

Child Process
Child PID: 868
Parent PID: 867
Executing ls using exec()
total 288
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi     0 Aug  8 04:16 File2.txt
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi   107 Aug 20 08:15 Makefile
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi    12 Aug 18 07:08 destination.txt
drwxr-xr-x 2 vishwa_teja_veerandi vishwa_teja_veerandi  4096 Aug  1 04:16 linux
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 15952 Jul 28 08:04 myprogram
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 16216 Aug 18 07:08 practical2.1
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi  1232 Aug 18 06:58 practical2.1.c
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi     0 Aug 18 07:00 practical2.1.txt
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 16296 Aug 18 06:34 practical3.1
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi  1069 Aug 11 06:28 practical3.1.c
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 16088 Aug 18 06:36 practical3.2
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi   357 Aug 11 06:32 practical3.2.c
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 16392 Aug 18 06:43 problem1
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi  1116 Aug  1 02:56 problem1.c
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi     0 Aug  1 04:17 problem1.txt
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi    25 Aug 20 06:53 sample.txt
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi    14 Aug 18 07:49 sample2.txt
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 15952 Aug 20 08:06 skill1.1
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi   140 Aug 20 08:04 skill1.1.c
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 16296 Aug 20 08:27 skill1.2
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi   730 Aug 20 08:25 skill1.2.c
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 16080 Aug 20 08:18 skill2
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi   341 Aug  8 03:34 skill2.c
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 16144 Aug  8 04:09 skill3
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi   377 Aug  8 03:37 skill3.c
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 16176 Aug  8 04:15 skill3main2
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi   478 Aug  8 03:39 skill3main2.c
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi    15 Aug  8 04:12 skill3main2.txt
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi     1 Aug 11 07:09 skill4.1.c
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi    12 Aug 19 09:56 source.txt
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 16216 Aug 20 08:18 sum1
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi   498 Aug  4 06:58 sum1.c
-rwxr-xr-x 1 vishwa_teja_veerandi vishwa_teja_veerandi 16472 Aug 20 08:20 task2
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi  1122 Aug  8 03:05 task2.c
drwxr-xr-x 2 vishwa_teja_veerandi vishwa_teja_veerandi  4096 Aug  1 04:13 teja
drwxr-xr-x 2 vishwa_teja_veerandi vishwa_teja_veerandi  4096 Jul 25 04:08 vishwa
-rw-r--r-- 1 vishwa_teja_veerandi vishwa_teja_veerandi    81 Jul 25 04:10 vishwa.c
Child process completed.
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ |
```

SESSION-2

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

A) Create a file by using

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano sum1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ |
```

Then press enter and the code terminal will open then write a code

```
  GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    printf("=== Interactive C Program ===\n");
    printf("Type 'exit' to quit the program.\n\n");

    while (1) {
        printf("Enter a command: ");
        fgets(input, sizeof(input), stdin);
        input[strcspn(input, "\n")] = '\0';
        if (strcmp(input, "exit") == 0) {
            printf("Exiting program...\n");
            break;
        }
        printf("You entered: %s\n", input);
    }

    return 0;
}
```

After writing the code save the your program by ctrl O and for coming back to command press ctrl X then compile it by the command

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano sum1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc sum1.c -o sum1
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ |
```

After compiling to get the output enter command

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano sum1.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc sum1.c -o sum1
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./sum1
=== Interactive C Program ===
Type 'exit' to quit the program.

Enter a command: ls
You entered: ls
Enter a command: pwd
You entered: pwd
Enter a command: exit
Exiting program...
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

Task 2: To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction.

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill1.3.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```

CODE:

```
  GNU nano 8.7.1
#include <stdio.h>
int main() {
char buffer[100];
int ch;
int index = 0;
printf("Type a command and press Enter:\n");
printf("input> ");
while ((ch = getchar()) != '\n' && ch != EOF) {
if (ch == 127 || ch == 8) {
if (index > 0) {
index --;
}
} else if (index < 99) {
buffer[index++] = ch;
}
}
buffer[index] = '\0';
printf("\nCommand entered: %s\n", buffer);
printf("Characters stored: %d\n", index);
return 0;
}
```

OUTPUT:

```
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ nano skill1.3.c
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ gcc skill1.3.c -o skill1.3
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$ ./skill1.3
Type a command and press Enter:
input> hello world

Command entered: hello world
Characters stored: 11
vishwa_teja_veerandi@LAPTOP-DID88SB3:~$
```